EXPERIMENT 7 – K-MEANS CUSTOMER CLUSTERING

AIM

To segment mall customers by Annual Income and Spending Score using K-Means clustering.

ALGORITHM

1. Load and inspect Mall_Customers.csv.
2. Select Annual Income and Spending Score and standardise them.
3. Use the Elbow Method for K=1 to 10.
4. Train K-Means using the chosen K and assign cluster labels.
5. Plot clusters and centroids; report counts and means; save labelled data.

In [9]

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```
import pandas as pd, matplotlib.pyplot as plt, seaborn as sns
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
import os

class="cm"># Automatically download the dataset if it's missing using a verifie
filename = class="s">"Mall_Customers.csv"
if not os.path.exists(filename):
    print(fclass="s">"{filename} not found. Downloading from GitHub...")
    url = class="s">"https://raw.githubusercontent.com/tirthajyoti/Machine-Lear
    try:
        df_temp = pd.read_csv(url)
        class="cm"># Ensure the column names match what the script expects
        df_temp = df_temp.rename(columns={
            class="s">'Annual Income (k$)': class="s">'Annual Income (k$)',
            class="s">'Spending Score (1-100)': class="s">'Spending Score (1-10
        })
```